HAMZA ALMUGHRABI

ML ENGINEER & DATA SCIENTIST & DATA ANALYST

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SUMMARY

Machine Learning & AI enthusiast with a solid foundation in **Data Science, Predictive Modeling**, and Deep Learning. Skilled in **Python**, **SQL**, and **Power BI**, with practical experience gained from projects focused on **data preprocessing**, **model development**, and **performance evaluation**. Passionate about applying ML techniques to solve real-world challenges, with a particular interest in Bissens

PROFESSIONAL SKILLS

Data Science & Architecture:

- Machine Learning (Regression, Classification, Clustering)
- Deep Learning (Neural Networks, CNN, RNN, NLP Basics)
- AI Agents & Workflow Automation (n8n Basics)
- Data Analytics & Modeling
- Data Architecture & Data Warehousing
- Data Cleaning, Transformation & ETL/ELT Pipelines
- Feature Engineering & Data Preprocessing
- Data Analytics, Reporting & Dashboards (Power BI, Matplotlib, Seaborn, Excel)

Programming & Databases:

- Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch basics)
- SQL (Query Optimization, MS Access)
- Cloud & Big Data: Databricks, GCP Basics (BigQuery)
- Web Technologies (HTML, CSS, JavaScript, React.js Basic, PHP Basic, Arduino, C++)

Tools & Productivity:

- GitHub, Jupyter Notebooks, Google Colab, VS Code
- Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Figma, Canva

EDUCATION

Intermediate Diploma in AI & Robotics Engineering (2022-2025)

Balqa'Applied University

- Machine Learning, Computer Vision, Data Structures.

CERTIFICATION

- Machine Learning (The Hope International)
- Deep Learning (The Hope International)
- Command Engineering for ai (Dubai future foundation)
- Preparing Data for Analysis with Microsoft Excel (Microsoft)
- Harnessing the Power of Data with Power BI (Microsoft)
- Extract, Transform and Load Data in Power BI (Microsoft)
- Data Modeling in Power BI (Microsoft)
- Data Analysis and Visualization with Power BI (Microsoft)
- Creative Designing in Power BI (Microsoft)
- Foundations: Data, Data, Everywhere (Google Cloud)
- Build Data Lakes and Data Warehouses (Google Cloud)
- Machine Learning with PyTorch (IBM Developer)
- Intermediate Python for Developers (DataCamp)
- Supervised Learning with scikit-learn (DataCamp)
- Web Design (Basic) (HTU)
- Amazon SageMaker AI Getting Started (AWS)

PROJECT EXPERIENCE

CAR INSURANCE CLAIM PREDICTION SYSTEM (2026 - Present)

Python, Scikit-Learn, Streamlit, Machine Learning | GitHub, Streamlit

- Developed an end-to-end machine learning system using an optimized **Soft Voting** Classifier (**Random Forest & Gradient Boosting**) with **GridSearchCV** to predict car insurance claims.
- Built and deployed an interactive **Streamlit web application** with real-time risk assessment, automated feature alignment, and modular project structure for production readiness.

TELECOM CUSTOMER CHURN ANALYSIS SYSTEM (2025)

Advanced Excel, Pivot Tables, Data Visualization | GitHub

- Processed and analyzed a telecom dataset of **6,687 customers** to identify key drivers behind a **26.86% churn rate** using advanced data aggregation and feature engineering.
- Developed an interactive **Excel dashboard** to segment churn by demographics (44% senior churn) and geography, capturing a critical **63.24% churn spike** in California.

DATA WAREHOUSE & ANALYTICS SYSTEM (2026)

SQL Server, Data Modeling | GitHub

- Designed a corporate **Data Warehouse** using **SQL Server** and **Databricks** (**Medallion Architecture**) to centralize ERP & CRM sales data.
- Executed robust ETL data cleaning and delivered SQL-based analytics to uncover critical insights into customer trends and product performance.

FINANCIAL RAG AI ASSISTANT SYSTEM (2026)

Python, LangChain, Streamlit, ChromaDB, Groq API, Llama 3.1-8b-instant | GitHub

- Deployed an end-to-end **RAG pipeline** to automate financial queries, optimizing data ingestion via parsing, dynamic chunking of PDFs, and **ChromaDB** vector storage.
- Integrated **Llama-3.1-8b-instant** via Groq API with low temperature ($temperature = 0.1$) to ensure strict precision, building an interactive **Streamlit UI** with source-verification tracking.

SALES FORECASTING SYSTEM (2025)

Python, Power BI, XGBoost | GitHub

- Built an **XGBoost** regression model ($R^2 = 0.999$, $RMSE = 56.6$) utilizing advanced feature engineering to forecast future sales trends.
- Developed an interactive **Power BI** dashboard to visualize sales drivers and deliver data-driven metrics for strategic business decisions.

WORKSHOP & JOB SIMULATION

- POWER BI_HR (Workshop)
- BCG X (Job Simulation)
- Your Real Guide to Entering (SHAI) (Workshop)

LANGUAGES

- Arabic (Native)
- English (B1-B2)